

α -labeling of banana tree graphs

Borna Gojšić

Faculty of Electrical Engineering and Computing, Zagreb, Croatia

borna.gojsic@fer.hr

This is joint work with Anamari Nakić

A banana tree graph, denoted by $B(n, k)$, is a tree constructed by connecting n star graphs S_k to a root vertex. A graceful labeling assigns the numbers 1 through $|E| + 1$ to vertices so that edge differences yield all values from 1 to $|E|$ exactly once; an α -labeling is a graceful labeling with an additional ordering property, motivated by work of Rosa [1], and is central in graph decomposition theory. This article builds upon previous research on graceful labelings of banana trees and explores their α -labelability. We utilize the symmetries of the $B(n, k)$ trees to give the number of all of the non-isomorphic α -labelings based on specific subsets of them which we call *proper* and *central* α -labelings. We classify all the α -labelings for all the $B(n, 1)$, $B(1, k)$ and $B(2, k)$ trees. We present α -labelings of the $B(n, k)$ tree, for $k \geq \frac{n}{2} + 1$ and $n \in \mathbb{N}$. Finally, we prove that the $B(n, k)$ tree is not α -labelable for $1 < k \leq \frac{n}{2}$ and any $n \geq 3$, classifying all the $B(n, k)$ trees based on their α -labelability.

References

- [1] A. Rosa, On certain valuations of the vertices of a graph, in: Theory of graphs (Internat. Sympos., Rome, 1966), Gordon & Breach, New York, 1967, 349–355.